

Alex Santora

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EDUCATION

Stevens Institute of Technology

Hoboken, NJ

B.E. Computer Engineering; B.S. Applied Mathematics

Sep. 2025 – May 2029

- **GPA: 3.88/4.00**; Dean's List; Edwin A. Stevens Scholarship; NCAA varsity baseball student-athlete.
- **Coursework:** Data Structures and Algorithms, Microprocessor Systems, Multi-variable Calculus, Linear Algebra

EXPERIENCE

University of New Hampshire

Durham, NH

Undergraduate Research Intern

May 2026 – Present

- Co-authored a paper introducing MDD-LPC, a privacy-preserving AI method that anonymizes speech by suppressing gender-identifying acoustic features while maintaining performance for emotion and pain detection.
- Achieved 33.6× faster anonymization and 20.24% lower power consumption than the GenGAN deep-learning baseline, reducing average processing latency from 17.50 seconds to 0.521 seconds.
- Evaluated privacy-utility tradeoffs across SVM, CNN, GRU-Mixer, and Wav2Vec2 machine-learning architectures using CREMA-D, RAVDESS, and TAME Pain datasets, reducing gender-inference F1 by 23.9–66.8% while preserving downstream clinical AI performance.

PROJECTS

Baseball Pitch Classification Model | *Python, Pandas, Scikit-Learn, Joblib, Pybaseball*

Jun. 2026

- Achieved 93.98% pitch-type classification accuracy and 0.94 weighted F1 on an 87,113-pitch test set by training a balanced Random Forest on 350K+ Statcast-style pitch records.
- Improved class separation for overlapping pitch profiles by engineering features from velocity, spin rate, spin axis, vertical/horizontal break, release point, and extension.
- Built a reproducible ML workflow by automating data ingestion, preprocessing, model training, persistence, and evaluation using pandas, scikit-learn, joblib, and pybaseball.

FutureWithAI.dev | *Founder & Developer*

Mar. 2026

- Founded an AI-literacy and mentorship platform that has helped 500+ students explore artificial intelligence, machine learning, and responsible AI.
- Led educational presentations at schools and provided personalized mentorship to help students develop technical skills and launch AI-focused projects.

Poker Neural Network | *Python, PyTorch, Reinforcement Learning, Neural Networks*

Jun. 2026

- Browser-based heads-up Texas Hold'em game with a neural network AI trained through reinforcement learning on 2M+ simulated hands.
- Improved AI decision quality by encoding 52-card deck state, hole cards, board cards, betting context, and hand-strength signals into neural-network-ready model inputs.

LEADERSHIP

NCAA Varsity Baseball

Stevens Institute of Technology

Student-Athlete; Leadership Council Member

Sep. 2025 – Present

- Balanced NCAA varsity competition with a rigorous engineering and applied mathematics course load by applying disciplined time management, consistent preparation, and team-first communication.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, Java, C#, C++, Arduino/C++

Machine Learning/Data: pandas, NumPy, scikit-learn, joblib, matplotlib, pybaseball, reinforcement learning, feature engineering, model evaluation

Engineering Tools: Git, SolidWorks, MQTT, ESP8266, ultrasonic sensing, embedded systems, browser-based applications